

used in industrial environments meet the challenges.

PoE IP surveillance on the bus. Security of public transportation is a rising concern for transportation companies to protect passengers and also for government authorities to confront terrorist attacks. One of the solutions is to deploy the IP surveillance system on the bus.

By using the DC power from the bus battery as power input, the PoE switch delivers power to the PoE IP video camera through the data cable. Without the additional power infrastructure and wiring, PoE is a better solution that fits into the limited storage space on a bus. However, due to the differences between the power systems of bus battery and IP camera, a PSE switch may be required to adapt to distinct implementation situations.

An additional DC-to-DC converter inserted between the bus battery and the PSE switch transforms the voltage from 24V to 48V for the PSE switch, so the PSE can apply suitable power level to the IP camera. A more straightforward approach is to adopt a PSE switch that is capable of 24V DC input and forwarding power to a 24V DC PD. This requires the PSE to accept a wider range of power inputs to provide a simpler and more effective solution when implementing PoE into most transportation systems.

PoE broadband data communication on trains. Wireless broadband data communication services in trains and other public transportation sys-

Industrial devices are often located outdoors or in control rooms, which could be extremely cold, hot or humid. So an industrial-grade PoE device should be able to operate under harsh conditions.



Fig. 5: Forced high-power PoE WiMax network

tems are becoming commonplace now. Passengers can simply use their laptop or PDA to browse the Internet, send and receive e-mails, or connect to a corporate data service when on the go. Train operators can also improve their service to customers; for instance, provide journey progress maps and entertainment network for audio/video streaming or multiplayer games.

The data communication service is achieved on board the train by deploying a Wi-Fi adaptor within each carriage and placing WiMax transceivers at each end of the train. The Wi-Fi and WiMax devices are powered by PoE switches in the carriage.

Enabling wireless data services by PoE eliminates the need to install and maintain power wiring. It saves cost and improves reliability. More and more transportation companies evaluate the PoE technology upon deploying wireless data communications. To provide seamless connection, the PoE switch should be reliable. In addition, the storage location for PoE devices is usually separated from passengers. The environment may be hot and humid because of no air-conditioning. Therefore the PoE device must be rugged, high-temperature-tolerant and equipped with anti-shock capability.

Forced high-power PoE WiMax

network. WiMax is an evolving telecom technology aimed at providing high-speed wireless data communication over long distances. It enables delivery of last-mile wireless broadband access in a cost-effective, flexible manner. In practice, the infrastructure is further simplified by using PoE-enabled WiMax devices.

Typically, a WiMax base station is composed of three base station units (BSUs). Each BSU has

a high-gain directional antenna that points to different directions in order to increase the coverage and throughput. To ensure reliability and long-range communication, the BSUs require higher power consumption, even up to 25W, and 48V DC power supply.

In this special case, a high-power-capable PoE switch is used to link four BSUs to the backbone network for data communication. Particularly, for these BSUs that do not support PD classification, each port connecting to the BSU is forced to deliver high power and limited to an upper bound to meet the ultra power consumption requirement as well as safety concerns.

To sum up

Industrial-grade PoE devices face diverse challenges from different power systems, including harsh operating environment and management requirement. By providing flexible installation and versatile management facilities, one can overcome deployment problems in industrial environments and bring the PoE technology into full play.

To be concluded next month...

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